

[TF 2.0] Dataset has no attribute 'make_one_shot_iterator' #29252

New issue

Closed BingyuZhou opened this issue on Jun 1, 2019 · 18 comments



BingyuZhou commented on Jun 1, 2019

Please make sure that this is a bug. As per our [GitHub Policy](#), we only address code/doc bugs, performance issues, feature requests and build/installation issues on [GitHub](#). tag:bug_template

System information

- Have I written custom code (as opposed to using a stock example script provided in TensorFlow): No
- OS Platform and Distribution (e.g., Linux Ubuntu 16.04): Google Colab
- TensorFlow version (use command below): 2.0
- Python version:
- Bazel version (if compiling from source):
- GCC/Compiler version (if compiling from source):
- CUDA/cuDNN version:
- GPU model and memory:

Describe the current behavior

Report attribute error as :

AttributeError: 'BatchDataset' object has no attribute 'make_one_shot_iterator'

Describe the expected behavior

Code to reproduce the issue

```
from __future__ import absolute_import, division, print_function, unicode_literals

# Install TensorFlow
!pip install tensorflow==2.0.0-alpha0

import tensorflow as tf
import tensorflow.keras.models as Models
import numpy as np

inc_dataset = tf.data.Dataset.range(100)
dec_dataset = tf.data.Dataset.range(0, -100, -1)
dataset = tf.data.Dataset.zip((inc_dataset, dec_dataset))
batched_dataset = dataset.batch(4)

iterator = batched_dataset.make_one_shot_iterator()
next_element = iterator.get_next()
```



Assignees

achandraa

Labels

compdata statawaiting response TF 2.0 typesupport

Projects

None yet

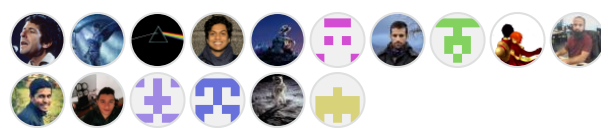
Milestone

No milestone

Development

No branches or pull requests

16 participants



srijithrajeev commented on Jun 3, 2019

As per [Release 2.0.0-alpha0](#), tf.data.Dataset.make_one_shot_iterator() has been deprecate in V1, removed from V2, and added to tf.compat.v1.data.make_one_shot_iterator().

👍 9



muddham self-assigned this on Jun 3, 2019



muddham added the TF 2.0 2.0.0-alpha0 compdata typesupport and removed TF 2.0 labels on Jun 3, 2019



muddham commented on Jun 3, 2019

@BingyuZhou Did you get chance to try as suggest by @srijithrajeev . I had tried and no error was shown. tf.compat.v1.data.make_one_shot_iterator(batched_dataset)

👍 1



muddham added the statawaiting response label on Jun 3, 2019



achandraa assigned achandraa and unassigned muddham on Jun 10, 2019



achandraa commented on Jun 10, 2019

Automatically closing due to lack of recent activity. Please update the issue when new information becomes available, and we will reopen the issue. Thanks!



achandraa closed this as completed on Jun 10, 2019



breadbread1984 commented on Sep 5, 2019

is there a counterpart of one_shot_iterator in tf 2.0. how can I keep the context of a dataset?

👍 26



fonnesbeck commented on Oct 21, 2019

Can someone address the question from @breadbread1984? There doesnt appear to be any clear guidance in the docs regarding what the best practices are in the absence of one_shot_iterator .

👍 29



gowthamkpr mentioned this issue on Oct 21, 2019

[TF 2.0] repeatdataset has no attribute 'make_one_shot_iterator' #33555

Closed



Giszy commented on Oct 25, 2019

```
import tensorflow as tf
import tensorflow.keras.models as Models
import numpy as np

inc_dataset = tf.data.Dataset.range(100)
dec_dataset = tf.data.Dataset.range(0, -100, -1)
dataset = tf.data.Dataset.zip((inc_dataset, dec_dataset))
batched_dataset = dataset.batch(4)

#iterator = batched_dataset.make_one_shot_iterator()
#next_element = iterator.get_next()

for next_element in batched_dataset:
    tf.print(next_element)
```

👍 26



breadbread1984 commented on Oct 26, 2019 • edited ↕

iterator is necessary when you use multiple trainset alternatively for training, for example, training mtcnn requires both wider face and celeba. you have to use iterator rather than a loop. but I have solved the problem buy tf.data.Dataset.iter()



abratichik commented on Oct 26, 2019

Hi @breadbread1984, could you please elaborate a bit how you managed to work around this issue? I'm trying to run the mnist.py from models/official/mnist and it fails in the eval_input_fn() for exact same reason - no make_one_shot_iterator() method.

Here is the code that fails:

```
def eval_input_fn():
    batched_dataset = dataset.test(flags_obj.data_dir).batch(flags_obj.batch_size)
    return batched_dataset.make_one_shot_iterator().get_next()
```



breadbread1984 commented on Oct 26, 2019 • edited ↕

@abratichik you can create iterator with

```
def eval_input_fn():
    batched_dataset = dataset.test(flags_obj.data_dir).batch(flags_obj.batch_size)
    return batched_dataset.__iter__()
```



you can get a batch with

```
iter = eval_input_fn();
sample = next(iter);
```



👍 5 🗨 5



viariasv commented on Nov 17, 2019

@abratichik you can create iterator with

```
def eval_input_fn():
    batched_dataset = dataset.test(flags_obj.data_dir).batch(flags_obj.batch_size)
    return batched_dataset.__iter__()
```

you can get a batch with

```
iter = eval_input_fn();
sample = next(iter);
```



thank you sir, that's the solution.

👍 2



skeller88 mentioned this issue on Jan 14, 2020

dataset.make_one_shot_iterator() raises AttributeError: 'MapDataset' object has no attribute 'make_one_shot_iterator' uber/petastorm#467

Open



hsh2438 commented on Jan 31, 2020

You just use iter() function like below.

```
iterator = iter(batched_dataset)
next_element = iterator.get_next()
```

👍 17 ❤ 5



maulikchevli mentioned this issue on Apr 2, 2020

TensorFlow V2 Bugs ialhashim/DenseDepth#110

Closed



willfenton mentioned this issue on Apr 13, 2020

Fixed bug in music VAE training script with make_one_shot_iterator magenta/magenta#1701

Merged



lvenugopalan added the TF 2.0 label on Apr 30, 2020

What is the solution for an error like this for below function?

```
def input_data():
    filenames = get_filenames()
    print(len(filenames))
    train_dataset = tf.data.Dataset.from_tensor_slices(filenames)
    train_dataset = train_dataset.shuffle(100).repeat()
    train_dataset = train_dataset.map(parse_image, num_parallel_calls=4).batch(batch_size)
    return train_dataset.make_one_shot_iterator()
```

Error is

```
Traceback (most recent call last):
  File "soft_n_cut_loss.py", line 307, in <module>
    iterator = input_data()
  File "C:\Users\Spring 2020\CSCE-B 657 Computer Vision\Project\unsupervised\unet2\U-Net\input_data.py", line 26, in
input_data
    return train_dataset.make_one_shot_iterator()
AttributeError: 'BatchDataset' object has no attribute 'make_one_shot_iterator'
```

If I use the iterator s mentioned in the comments above then I get error as:

```
Traceback (most recent call last):
  File "soft_n_cut_loss.py", line 307, in <module>
    iterator = input_data()
  File "C:\Users\Spring 2020\CSCE-B 657 Computer Vision\Project\unsupervised\unet2\U-Net\input_data.py", line 27, in
input_data
    iterator = iter(train_dataset)
  File "C:\ProgramData\Anaconda3\lib\site-packages\tensorflow_core\python\dataops\dataset_ops.py", line 334, in
__iter__
    raise RuntimeError("__iter__() is only supported inside of tf.function ")
RuntimeError: __iter__() is only supported inside of tf.function or when eager execution is enabled.
```

Can you please help me, how I can progress in this case? It would be great if someone can help me with the change I need to make in my function so that it works

👍 2

kvijaysai commented on May 3, 2020

Hey,

It's working for me. I did below 2 things:

1. Added this in import libraries, earlier it was import tensorflow as tf

```
import tensorflow.compat.v1 as tf
tf.disable_v2_behavior()
```

2. In main function, added the below snippet

```
iterator = input_data()
images = iterator.get_next()
```

👍 1

zohaibmohammad commented on Aug 19, 2020

Hello,

This solves my error.

```
# import tensorflow as tf
import tensorflow.compat.v1 as tf
tf.disable_v2_behavior()
```

```
iterator = tf.compat.v1.data.make_one_shot_iterator(dataset)
```

👍 1

Harrypotterrrr commented on Oct 1, 2020

@kvijaysai Hi, downgrading the v2 to fit the compatibility is not a good workaround to deal with such kind of things. You can try the following code

```
def input_data():
    filenames = get_filenames()
    print(len(filenames))
    train_dataset = tf.data.Dataset.from_tensor_slices(filenames)
    train_dataset = train_dataset.shuffle(100).repeat()
    train_dataset = train_dataset.map(parse_image, num_parallel_calls=4).batch(batch_size)
    return train_dataset
train_iter = iter(train_dataset)
for i in range(steps*num):
    input, output = train_iter.get_next()
```

You could check [tf.data iterator](#) for more info.

Harrypotterrrr commented on Oct 1, 2020 • edited

I have got a similar problem. I deploy my trainingdataset and validation dataset on multi GPUs. for the training dataset, I use the following code to build the dataset, while for the validation dataset, that requires me to iterate for each time of specified steps. Thus, I choose tf.data.iterator to deal with such things, as the above comment shows. However, when I tried to deploy the val_iter to multi gpus, the error is thrown. Anyone could help me out with this. the error is "QueueIterator object has no attribute '_variant_tensor'" which points to the last line of the following code. Note that I use iter to make val_dataset iterable. but the TF2 forbids me to do that.

```
with self.mirrored_strategy.scope():
    self.model = model(config)
    self.optimizer = tf.optimizers.Adam(learning_rate=self.config["lr"])
    self.train_dataset, self.val_dataset, self.test_dataset = dataloader()
    self.train_dataset = self.mirrored_strategy.experimental_distribute_dataset(self.train_dataset)
    self.val_dataset = self.mirrored_strategy.experimental_distribute_dataset(iter(self.val_dataset))
```

btw, my TF version is the latest 2.3

boobootoo2 mentioned this issue on Feb 27, 2021

AttributeError: 'PrefetchDataset' object has no attribute 'make_one_shot_iterator'
salu133445/musegan#115

flackbash mentioned this issue on Feb 22, 2022

Code does not run with tensorflow v2.5.0 dalab/end2end_neural_el#38

soran-ghaderi commented on Apr 12, 2022 • edited

You can create an iterator using the following function (Tensorflow 2x):

```
def dataset_iterator(output_types, output_shapes):
    dataset = tf.data.Dataset.from_generator(self._training_data_generator,
                                             output_types=output_types,
                                             output_shapes=output_shapes)
    dataset = dataset.repeat().prefetch(prefetch_batches)
    dataset_iterator = dataset.__iter__()
    return dataset_iterator
```

wzZero commented on Apr 12, 2022

just return iterator is feasible, but im not sure how to make sure that self_training_data_generator is right,do i need to write a for-loop and yield something? here's part of my code:

```
def individual_batch_generator(self,inputs,targets):
    assert len(inputs) == len(targets)
    indices = np.arange(len(inputs))
    np.random.shuffle(indices)
    for index in range(0,len(inputs)):
        if self.in_memory_data:
            yield inputs[index], targets[index]
        else:
            input_features = self.individual_load_feature(inputs[index])
            yield input_features, targets[index]
```

and when i run my code it still errors with input ran out of data, i just put the returned iterator into model.fit T.T